

SHM Worked Example 5.16

Reference: Applied Mechanics: Reed's Vol. 2

Ex. 5.16: A machine component of mass $m = 2.25$ kg moves with simple harmonic motion. The amplitude is $A = 380$ mm, and the frequency of oscillation is 120 oscillations per minute. Determine:

1. The maximum accelerating force.
2. The accelerating force when the displacement is 250 mm from the mean position.

Given Data

- Mass, $m = 2.25$ kg
- Amplitude, $A = 380$ mm = 0.38 m
- Frequency, $f = \frac{120}{60} = 2$ Hz

The angular velocity of the motion is given by:

$$\omega = 2\pi f = 2\pi \times 2 = 4\pi = 12.5664 \text{ rad/s}$$

(i) Maximum Accelerating Force

The maximum acceleration occurs at maximum displacement ($x = A$) and is given by:

$$a_{\max} = \omega^2 A$$

Substituting the values:

$$a_{\max} = (4\pi)^2 \times 0.38 = 16\pi^2 \times 0.38 = 60.0072 \text{ m/s}^2$$

The maximum accelerating force is:

$$F_{\max} = m a_{\max} = 2.25 \times 60.0072 = 135.0162 \text{ N}$$

(ii) Accelerating Force at $x = 250$ mm

The acceleration at displacement x is given by:

$$a = \omega^2 x$$

Substituting $x = 250$ mm = 0.25 m:

$$a = (4\pi)^2 \times 0.25 = 16\pi^2 \times 0.25 = 39.4784 \text{ m/s}^2$$

The accelerating force is:

$$F = m a = 2.25 \times 39.4784 = 88.8264 \text{ N}$$

Final Answers

1. Maximum accelerating force: $F_{\text{max}} = 135.0162 \text{ N}$
2. Accelerating force at $x = 250$ mm: $F = 88.8264 \text{ N}$